

Alexander B. Alleman

Assistant Professor of Chemistry | Western Colorado University | Natural and Environmental Sciences Department
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ACADEMIC POSITION

2025 - Assistant Professor of Chemistry
Natural and Environmental Sciences Department
Western Colorado University

EDUCATION

2022-2025 Postdoctoral Fellowship
Biological Sciences, University of Idaho
Postdoctoral Advisor: Dr. Christopher J. Marx

2017-2021 Ph.D. in Molecular Plant Science
Institute of Biological Chemistry, Washington State University
Doctoral Advisor: Dr. John W. Peters

2009-2014 B.A. Biochemistry and Cell Biology
Western Colorado University

TEACHING EXPERIENCE

2025- Instructor at Western Colorado University
CHEM 111: General Chemistry I
CHEM 112: General Chemistry I Lab
CHEM 234: Intro to Organic and Biochemistry Lab
CHEM 471: Biochemistry I
CHEM 113: General Chemistry II
CHEM 472: Biochemistry II
CHEM 474: Biochemistry Lab

2022-2025 Instructor at University of Idaho
BIOL 300: Introduction to Biochemistry
CNR 400: Special Topics EMBER

2022-2025 Guest lecturer at University of Idaho
BIOL 504: Microbial Physiology
BIOL 450: Microbiomes

2015-2016 Teaching assistant at Montana State University
CHMY 321: Organic Chemistry I
CHMY 323: Organic Chemistry II

PUBLICATIONS (**, direct mentee)

Alexander B. Alleman, Akorede L Seriki, Dawson J. Mathes, Monica J. Pedroni, Nkrumah A. Grant, Christopher J. Marx. 2026. Evolutionary Integration of a Foreign Aromatic Catabolic Pathway Drives Metabolic Trade-offs in *Methylobacterium extorquens* PA1. BioRxiv doi:

Keaton J. Adams**, Christopher J. Marx, **Alexander B. Alleman**. 2025. Storage polymer polyhydroxybutyrate promotes recovery and limits phenotypic heterogeneity in regrowth from carbon starvation in *Methylobacterium extorquens*. BioRxiv.doi: 10.64898/2025.12.12.690574

Akorede L. Seriki, **Alexander B. Alleman**, Tomislav Ticak, Alyssa C. Baugh, Jack W. Creagh, Christopher J. Marx. 2025. Engineered *Methylobacterium extorquens* grows well on methoxylated aromatics due to its formaldehyde metabolism and stress response. Msphere doi: 10.1128/msphere.00171-25

Sabrina J. Tuson, Shahla J. Nemati, **Alexander B. Alleman**, Christopher J. Marx, Andreas E. Vasdekis. 2025. Microbial Cell-Adaptive Segmentation in Quantitative Phase Imaging. BioRxiv doi: 10.1101/2025.11.13.688197

Alexander B. Alleman, Sergey Stolyar, Christopher J Marx, Jean-Baptiste Leducq. 2025. Led astray by 16S rRNA: *Methylobacterium* and reinstatement of the genus *Methylobacterium* using phylogenomics and metagenomics. ISME Journal. doi: 10.1093/ismejo/wraf011

Alexander B. Alleman, John W. Peters. 2023. Mechanisms for generating low potential electrons across the metabolic diversity of nitrogen-fixing bacteria. Applied and Environmental Microbiology. doi: 10.1128/aem.00378-23

Alexander B. Alleman, Florence Mus, Amaya Garcia Costas, John W. Peters. 2022. Rnf and Fix Have Specific Roles during Aerobic Nitrogen Fixation in *Azotobacter vinelandii*. Applied and Environmental Microbiology. doi: 10.1128/aem.01049-22

Alexander B. Alleman, Florence Mus, John W. Peters. 2021. Metabolic Model of the Nitrogen-Fixing Obligate Aerobe *Azotobacter vinelandii* Predicts Its Adaptation to Oxygen Concentration and Metal Availability. Mbio. doi: 10.1128/mBio.02593-21

Natasha Pence, Nathan Lewis, **Alexander B. Alleman**, Lance C. Seefeldt, and John W. Peters. 2021. Revealing a role for the gamma subunit in mediating interactions between the nitrogenase component proteins. Journal of Inorganic Biochemistry. 214:111273. doi: 10.1016/j.jinorgbio.2020.111273.

Alexander B Alleman, Yesuf Assen Mohammed, Kent A McVay, Qasim A Khan, Patrick Carr, John Miller, Zachariah Miller, Jessica Torrior, Peggy Lamb, Florence Mus, Chengci Chen, and John W Peters. 2021. Drivers of Bacterial Community Structure and Co-occurrence in a Northern Great Plains Pulse Crop Rotation System. Applied Soil Ecology. 157, 103737. doi: 10.1016/j.apsoil.2020.103737

Florence Mus, Hsin-Hua Wu, **Alexander B. Alleman**, Krista A Shisler, Oleg A. Zadvornyy, Brian Bothner, Jennifer L. DuBois, and John W. Peters. 2020. Insights into unique carboxylation reactions in the metabolism of propylene and acetone. Biochemical Journal. 477(11):2027-2038. doi: 10.1042/BCJ20200174

Alexander B. Alleman, Gregory A. Prussia, Florence Mus, and John W. Peters. 2019. Acetone Carboxylase. Encyclopedia of Inorganic and Bioinorganic Chemistry. doi: 10.1002/9781119951438.eibc2701

Florence Mus, **Alexander B. Alleman**, Natasha Pence, Lance C. Seefeldt, John W. Peters. 2018. Exploring the alternatives of biological nitrogen fixation. Metallomics. 10(4):523-538. doi: 10.1039/c8mt00038g

Florence Mus, Brian J. Eilers, **Alexander B. Alleman**, Burak V. Kabasakal, James W. Murray, Boguslaw P. Nocek, Jennifer L. DuBois, John W. Peters. 2017. Mechanism of ATP-Dependent Acetone Carboxylation. Scientific Reports. 7(1):7234. doi: 10.1038/s41598-017-06973-8

ORAL PRESENTATIONS

The tortoise and the hare: Eco-evolutionary dynamics of a synthetic microbial community
Western Colorado University NES seminar, Gunnison, CO February 2026

When Bacteria Resist Change: The Challenge of Altering Central Metabolic Pathways
Science on Tap, Outreach talk for the public, Moscow, ID November 2024

Using experimental evolution to reduce the burden of formaldehyde stress while consuming methoxylated aromatics with an engineered *Methylobacterium extorquens* strain.
Symposium on Biomaterials, Fuels and Chemicals, Portland, OR April 2023

The effects of oxygen limitations on C1 metabolism in *Methylobacterium extorquens* PA1.
Northwest ASM meeting, Spokane, WA November 2022

Balancing oxygen and nitrogen reduction: how *Azotobacter vinelandii* fixes nitrogen under high aeration conditions.
Molecular Plant Science Seminar Series, Pullman, WA November 2020

Bioenergetics of nitrogen fixation: Remodeling energy metabolism for diazotrophic growth.
Molecular Plant Science Seminar Series, Pullman, WA September 2019

POSTER PRESENTATIONS

GRC-Molecular Basis of Microbial One Carbon Metabolism August 2024
Waterville Valley, NH, USA

DOE Basic Energy Science PI meeting March 2024
Washington D.C., USA

GRC-Microbial Population Biology <i>Andover, NH, USA</i>	July 2023
DOE Basic Energy Science PI meeting <i>Washington D.C., USA</i>	April 2023
GRC-Molecular Basis of Microbial One Carbon Metabolism <i>Southbridge, MA, USA</i>	August 2022
ASM Microbe <i>Online</i>	June 2021
Conference on Constraint-Based Reconstruction and Analysis (COBRA) <i>Online</i>	March 2021
Multiscale Microbial Dynamics Modeling Summer School <i>Online</i>	July 2020
Cluster of Excellence on Plant Sciences Summer School <i>Cologne, Germany</i>	May 2019
Awarded best poster	
Western Photosynthesis Conference <i>Friday Harbor, WA, USA</i>	January 2019
European Nitrogen Fixation Conference <i>Stockholm, Sweden</i>	August 2018
West Coast Protein Crystallography Workshop <i>Asilomar, CA, USA</i>	March 2017
Awarded best poster	
Northwest Crystallography Workshop <i>Pullman, WA, USA</i>	June 2016

TRAINING EXPERIENCE

Undergraduate research

Noah Arts (2024 – present, current undergraduate U of I)
 Keaton Adams (2023 – present, current undergraduate at U of I)
 Grace Black (Summer 2023, Post-Bac, Pacific Northwest National Laboratory)
 Dawson Mathes (2022-2024, Scientist I, Evotec Biologics)
 Aasiya Islam (2019, UW Medical School)

HONORS AND AWARDS

John & Maggie McDougall Scholarship	2020
Clarence “Bud” Ryan Institute of Biological Chemistry Travel Scholarship	2019
Thornton Biology Research Fellowship	2013

SCIENCE PROFESSIONAL ASSOCIATIONS

American Society of Microbiology	2018-present
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ACADEMIC SERVICE

Journal Reviewer – Journal of Bacteriology, Microbiology Spectrum, mSystems, Molecular Biology and Evolution, Cell Systems

Grant Reviewer- Department of Energy - FAIR

COMMUNITY OUTREACH

Methylothon	2021-2023
Hands-on experiments of isolating leaf bacteria and demonstrating evolution to middle and high school students	
Skype a Scientist	2020-2022
Give lectures and Q&A sessions for classrooms around the country	
Plant Science Day	2018- 2020
An annual event that demonstrates plant science for children of all ages	